Домашнее задание № 9

1. Берем тестовый датасет

```sql
CREATE TABLE covid19 (
    date Date,
    location_key LowCardinality(String),
    new_confirmed Int32,
    new_deceased Int32,
    new_recovered Int32,
    new_tested Int32,
    cumulative_confirmed Int32,
    cumulative_deceased Int32,
    cumulative_recovered Int32,
    cumulative_tested Int32
)
ENGINE = MergeTree
ORDER BY (location_key, date);
```

2. Преобразуем в реплицируемую таблицу, используя макрос replica

```sql
-- Создаем реплицируемую таблицу
CREATE TABLE covid19 (
    date Date,
    location_key LowCardinality(String),
    new_confirmed Int32,
    new_deceased Int32,
    new_recovered Int32,
    new_tested Int32,
    cumulative_confirmed Int32,
    cumulative_deceased Int32,
    cumulative_recovered Int32,
    cumulative_tested Int32
)
ENGINE = ReplicatedMergeTree(
    '/clickhouse/tables/{shard}/covid19',
    '{replica}'
)
ORDER BY (location_key, date);
```

3. Добавляем 2 реплики

Макросы для реплик

replica-1-macros.xml – Блокнот

Файл Правка Формат Вид Справка

```xml
<clickhouse>
    <macros>
        <shard>1</shard>
        <replica>replica-1</replica>
    </macros>
</clickhouse>
```

replica-2-macros.xml – Блокнот

Файл Правка Формат Вид Справка

```xml
<clickhouse>
    <macros>
        <shard>1</shard>
        <replica>replica-2</replica>
    </macros>
</clickhouse>
```

Фрагменты docker-compose.yml

docker-compose.yml – Блокнот

Файл Правка Формат Вид Справка

```yaml
version: '3.8'

services:
  # Сервис координации (обязателен для репликации)
  zookeeper:
    image: zookeeper:3.8
    container_name: zookeeper
    hostname: zookeeper
    restart: unless-stopped
    ports:
      - "2181:2181"
    environment:
      ZOO_MY_ID: 1
      ZOO_SERVERS: server.1=zookeeper:2888:3888;2181
    networks:
      - clickhouse-network

  # Первая реплика ClickHouse
  clickhouse-replica-1:
    image: clickhouse/clickhouse-server:latest
    container_name: clickhouse-replica-1
    hostname: clickhouse-replica-1
    restart: unless-stopped

    ports:
      - "8123:8123"      # HTTP
      - "9000:9000"      # TCP
      - "9009:9009"      # interserver
```

```yaml
    volumes:
      - ./config/listen.xml:/etc/clickhouse-server/config.d/listen.xml
      - ./config/optimized_config.xml:/etc/clickhouse-server/config.d/optimized_config.xml
      - ./config/replica-1-macros.xml:/etc/clickhouse-server/config.d/macros.xml  # НОВОЕ
      - ./config/keeper.xml:/etc/clickhouse-server/config.d/keeper.xml            # НОВОЕ
      - ./user_files/:/var/lib/clickhouse/user_files/
      - clickhouse-data-1:/var/lib/clickhouse
      - clickhouse-logs-1:/var/log/clickhouse-server

    ulimits:
      nofile:
        soft: 262144
        hard: 262144

    depends_on:
      - zookeeper

    networks:
      - clickhouse-network

  # Вторая реплика ClickHouse
  clickhouse-replica-2:
    image: clickhouse/clickhouse-server:latest
    container_name: clickhouse-replica-2
    hostname: clickhouse-replica-2
    restart: unless-stopped

    ports:
      - "8124:8123"      # HTTP (другой порт)
      - "9001:9000"      # TCP (другой порт)
      - "9010:9009"      # interserver (другой порт)


    volumes:
      - ./config/listen.xml:/etc/clickhouse-server/config.d/listen.xml
      - ./config/optimized_config.xml:/etc/clickhouse-server/config.d/optimized_config.xml
      - ./config/replica-2-macros.xml:/etc/clickhouse-server/config.d/macros.xml  # НОВОЕ
      - ./config/keeper.xml:/etc/clickhouse-server/config.d/keeper.xml            # НОВОЕ
      - ./user_files/:/var/lib/clickhouse/user_files/
      - clickhouse-data-2:/var/lib/clickhouse
      - clickhouse-logs-2:/var/log/clickhouse-server

    ulimits:
      nofile:
        soft: 262144
        hard: 262144

    depends_on:
      - zookeeper

    networks:
      - clickhouse-network

volumes:
  clickhouse-data-1:
    driver: local
  clickhouse-data-2:
    driver: local
  clickhouse-logs-1:
    driver: local
  clickhouse-logs-2:
    driver: local
```

4. Создаем таблицу на каждой реплике и заполняем данными на одной реплике, на второй они появляются автоматически

```sql
-- Создаем реплицируемую таблицу на каждой реплике
CREATE TABLE covid19 (
    date Date,
    location_key LowCardinality(String),
    new_confirmed Int32,
    new_deceased Int32,
    new_recovered Int32,
    new_tested Int32,
    cumulative_confirmed Int32,
    cumulative_deceased Int32,
    cumulative_recovered Int32,
    cumulative_tested Int32
)
ENGINE = ReplicatedMergeTree(
    '/clickhouse/tables/{shard}/covid19',
    '{replica}'
)
ORDER BY (location_key, date);

-- вставляем данные на одной реплике
INSERT INTO covid19
   SELECT *
   FROM
      url(
        'https://storage.googleapis.com/covid19-open-data/v3/epidemiology.csv',
        CSVWithNames,
        'date Date,
        location_key LowCardinality(String),
        new_confirmed Int32,
        new_deceased Int32,
        new_recovered Int32,
        new_tested Int32,
        cumulative_confirmed Int32,
        cumulative_deceased Int32,
        cumulative_recovered Int32,
        cumulative_tested Int32'
    );
```

5. Выполняем запросы

```sql
-- выполнить 1 раз
SELECT
    getMacro('replica') AS current_replica,
    *
FROM remote('clickhouse-replica-1,clickhouse-replica-2', system.parts, 'default', 'clickme')
FORMAT JSONEachRow;
```

Результат 1

SELECT getMacro('replica') AS current_replica, * F | Введите SQL выражение чтобы отфильтровать результаты

	current_replica	partition	name	uuid	part_type	active	marks	rows
1	replica-2	tuple()	all_0_6_1	00000000-0000-0000-0000-000000000000	Wide	1	952	7 783 670
2	replica-2	tuple()	all_7_7_0	00000000-0000-0000-0000-000000000000	Wide	1	137	1 111 953
3	replica-2	tuple()	all_8_8_0	00000000-0000-0000-0000-000000000000	Wide	1	137	1 111 953
4	replica-2	tuple()	all_9_9_0	00000000-0000-0000-0000-000000000000	Wide	1	137	1 111 953
5	replica-2	tuple()	all_10_10_0	00000000-0000-0000-0000-000000000000	Wide	1	137	1 111 953
6	replica-2	tuple()	all_11_11_0	00000000-0000-0000-0000-000000000000	Wide	1	37	294 343
7	replica-2	202608	202608_1_47_11	00000000-0000-0000-0000-000000000000	Compact	0	2	568
8	replica-2	202608	202608_1_51_12	00000000-0000-0000-0000-000000000000	Compact	1	2	592

Файл **replicas_parts.json**

```sql
-- на каждой реплике
SELECT * FROM system.replicas
FORMAT JSONEachRow;
```

plicas 1

ELECT * FROM system.replicas | Введите SQL выражение чтобы отфильтровать результаты

	database	table	uuid	engine	is_leader	can_become_leader	is_readonly
1	myanalytics	covid19	db57b1b0-b4c4-43bc-ac73-07d2a35e832d	ReplicatedMergeTree	1	1	0

```sql
-- на каждой реплике
SELECT * FROM system.replicas
FORMAT JSONEachRow;
```

replicas 1

SELECT * FROM system.replicas Введите SQL выражение чтобы отфильтровать результаты

	database	table	uuid	engine	is_leader	can_become_leader	is_readonly	
1	myanalytics	covid19	171e34f8-e277-4c32-a7ac-3270dd5cda94	ReplicatedMergeTree	1	1	0	[NU

Файлы **replica1.json** и **replica2.json**

6. Добавление TTL

```sql
-- на каждой реплике
ALTER TABLE covid19 MODIFY TTL date + INTERVAL 7 DAY;
```

7. Проверка результата работы

```sql
-- проверка результата
SHOW CREATE TABLE covid19;

CREATE TABLE myanalytics.covid19
(
    `date` Date,
    `location_key` LowCardinality(String),
    `new_confirmed` Int32,
    `new_deceased` Int32,
    `new_recovered` Int32,
    `new_tested` Int32,
    `cumulative_confirmed` Int32,
    `cumulative_deceased` Int32,
    `cumulative_recovered` Int32,
    `cumulative_tested` Int32
)
ENGINE = ReplicatedMergeTree('/clickhouse/tables/{shard}/covid19', '{replica}')
ORDER BY (location_key, date)
TTL date + toIntervalDay(7)
SETTINGS index_granularity = 8192;
```